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PAPER_1001: SMBH Binary Merger F_U_Bi with Phonon Damping

Abstract

We compute $F_{U,Bi}$ for SMBH binary inspiral systems where phonon damping modifies the merger waveform. The SCm buoyancy field introduces a frequency-dependent damping envelope via $S_{26}^{(3)}$.

1. Core Equation

$$F_{\{U,Bi\}}^{\{binary\}} = \rho_{SCm} \cdot V \cdot S_{26}^{(3)2} \cdot ratio \cdot (1 + \Phi(\Gamma) \cdot e^{-\Gamma/\Gamma_0})$$

The damping envelope suppresses high-frequency phonon modes during the final inspiral.

2. Results

For a $5.5 \times 10^7 M_\odot$ primary: $F_{U,Bi} = 5.47 \times 10^{35} \text{ m/s}^2$ with $S_{26}^{(3)}$ -calibrated buoyancy correction.

3. Implementation

File: fubi_agn_ns_mergers.py, class SMBHBinaryMergerFUBiCalc. CP4 class #585.